

Calibration of Deep Neural Networks under Data-Scarce and Decision-Critical Conditions

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Abstract

Post-hoc calibration methods improve confidence reliability in deep learning models deployed in safety-critical settings. However, their behavior under limited calibration data and decision-critical reliability requirements remains insufficiently studied. We analyze Temperature Scaling, Entropy-based Temperature Scaling and Energy-based Calibration under varying calibration-set sizes and region-restricted evaluation metrics. The study provides a systematic comparison of calibration robustness under data scarcity and high-confidence decision regimes, supporting method selection in safety-relevant applications.

1 Introduction

Deep neural networks often produce probability estimates that deviate from true error likelihoods [9]. Predicted confidences may therefore overestimate actual correctness, misleading decision-makers.

Reliable confidence is crucial in safety-critical settings, where overconfident errors can have severe consequences [4]. In applications such as medical diagnosis or autonomous driving, actions are often triggered only when predicted confidence exceeds a high threshold, making miscalibration in these regions harmful. Even advanced uncertainty estimation approaches, including deep ensembles, exhibit calibration errors under realistic conditions [11], motivating the need for robust calibration methods beyond idealized benchmarks.

The reviewed literature indicates two practically relevant gaps: (1) **data-scarce calibration**, where only a few labeled samples are available for calibration, and (2) **decision-critical calibration**, where calibration quality matters in probability regions that directly influence subsequent decisions.

Under data-scarce calibration, many post-hoc methods exhibit performance degradation due to increased parameter variance and overfitting [1, 10].

Decision-critical calibration focuses on probability intervals around predefined confidence thresholds, where prediction errors have higher operational impact. In such regions, local calibration behavior becomes particularly important. Global metrics such as Expected Calibration Error (ECE) do not explicitly capture calibration performance in these decision-relevant intervals [2, 15]. Unlike classical cost-sensitive learning, this notion emphasizes calibration near confidence thresholds rather than explicit cost matrices.

While calibration under distribution shift is well studied [13], the combined impact of limited calibration data and decision-critical evaluation remains largely unexplored. This work addresses the gap by analyzing calibration under both conditions. We evaluate representative post-hoc calibration methods using global and region-restricted metrics. The study provides guidance for selecting calibration techniques in data-scarce, decision-critical scenarios such as clinical screening or safety monitoring applications.

2 Related Work

Post-hoc calibration research includes global scaling methods, adaptive approaches for limited calibration data and decision-focused evaluation strategies. Representative works addressing these aspects are summarized in Table 1.

Table 1: Coverage of data-scarce and decision-critical calibration in related work.

Key Paper	Data-Scarce Calibration	Decision-Critical Calibration
Mozafari et al. [10]	(✓)	×
Balanyà et al. [1]	(✓)	×
Yang et al. [15]	×	✓
Canal et al. [2]	×	✓
This work	✓	✓

✓: explicitly addressed (✓): partially or indirectly addressed ×: not addressed.

Among post-hoc calibration methods, Temperature Scaling (TS) [5] serves as a widely adopted baseline, applying a single global scaling parameter to logits. However, global scaling may be insufficient for modern networks [9, 11]. Under limited calibration data, global methods may overfit. Adaptive approaches incorporate instance-level uncertainty signals to improve robustness. For instance, Entropy-based Temperature Scaling (ETS) adapts the temperature as a function of predictive entropy [1], while attended calibration mechanisms dynamically adjust confidence based on feature representations [10]. Recent work observes that global metrics may obscure calibration behavior in critical confidence regions. Decision-driven or cost-aware calibration research emphasizes evaluation or optimization in predefined probability intervals relevant for downstream decisions [2, 8, 15]. However, robustness under limited calibration data is rarely examined.

3 Methods

This section introduces the post-hoc calibration methods evaluated in this study. Each method is described in terms of its formulation, parameterization and underlying calibration principle. Let $\mathbf{z} \in \mathbb{R}^K$ denote the vector of logits produced by the trained classifier for an input sample, where K is the number of classes. Calibration parameters are learned on a held-out subset by minimizing task-specific Negative Log-Likelihood. For evaluation, all methods are converted to correctness-based confidence estimates to ensure comparability.

Temperature Scaling. TS is a post-hoc calibration method that rescales the logits of a trained classifier using a single scalar parameter $T > 0$ [5]. The calibrated class probabilities are obtained as

$$p_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}, \quad (1)$$

where z_i denotes the logit for class i . The temperature parameter T is learned on the calibration subset. TS preserves class labels while globally adjusting confidence.

Entropy-based Temperature Scaling. ETS extends TS by using an instance-dependent temperature based on predictive uncertainty [1]. Let

$$p = \text{softmax}(\mathbf{z}) \quad (2)$$

and define the normalized predictive entropy as

$$H(\mathbf{z}) = -\frac{1}{\log K} \sum_{i=1}^K p_i \log p_i. \quad (3)$$

In the multi-label case, entropy is computed per label using the sigmoid probability.

ETS models the temperature as a function of entropy,

$$T(\mathbf{z}) = \exp(a H(\mathbf{z}) + b), \quad (4)$$

where a and b are scalar parameters learned from the calibration data. The exponential ensures $T(\mathbf{z}) > 0$. Calibrated probabilities are then computed using equation 1 with the instance-dependent temperature.

Energy-based Calibration. Energy-based Calibration derives confidence estimates from an energy score computed over the logits rather than directly using softmax probabilities [6]. For multi-class classification, the energy of an input is defined as

$$E(\mathbf{x}) = -\log \sum_{j=1}^K \exp(z_j). \quad (5)$$

In the multi-label setting, we adapt the energy formulation by computing it independently per label, yielding a label-wise energy score consistent with sigmoid-based outputs. Instead of using softmax confidence, a monotonic mapping function $f(E)$ is learned to convert energy values into calibrated confidence estimates. In this work, $f(E)$ is implemented as logistic regression on the negative energy,

$$\hat{p} = \sigma(a \cdot (-E(\mathbf{x})) + b), \quad (6)$$

where $\sigma(\cdot)$ denotes the sigmoid function and a, b are scalar parameters. Here, $\hat{p}$ is interpreted as the estimated probability that the predicted class label is correct, i.e., a scalar correctness confidence.

4 Experimental Apparatus

This section outlines the experimental setup used to evaluate calibration performance under data-scarce and decision-critical conditions, including datasets, preprocessing, training procedure, calibration protocol and evaluation metrics.

4.1 Datasets

Experiments are conducted on two safety-critical image classification datasets from distinct domains: ChestX-ray14 [14] for thoracic disease classification and BDD100K [17] for traffic scene classification. ChestX-ray14 is treated as a multi-label classification task, while BDD100K is formulated as a multi-class classification task. For ChestX-ray14, the official patient-level split is used, while BDD100K follows the standard 70K/10K/20K train/validation/test split. Detailed statistics are provided in Appendix A.1. Calibration samples are drawn exclusively from the validation sets, while test data remain unseen until final evaluation. To simulate data scarcity, multiple calibration subsets of predefined sizes $n_{\text{cal}} \in \{50, 100, 250, 500, 1000, 5000\}$ are sampled from the validation set. Stratified sampling is performed with respect to the ground-truth class labels to preserve the original

class distribution within each subset and avoid distortions due to class imbalance.

4.2 Pre-processing

Images are resized to a fixed resolution of 224×224 pixels to ensure architectural consistency and comparability across datasets and backbone models. This resolution follows the standard ImageNet pretraining configuration of ConvNeXt-Tiny [7] and ViT-B/16 [3], avoiding additional confounding factors such as resolution-dependent calibration behavior. During training, data augmentation includes random cropping and horizontal flipping following standard image classification practice [16]. These transformations increase robustness to small geometric variations without altering semantic content and have been applied in prior ChestX-ray classification work [12]. Identical preprocessing is applied to both datasets and across calibration methods to ensure fair and comparable calibration evaluation.

4.3 Procedure

For each dataset, images are preprocessed as described above and used to train two modern backbone architectures: ConvNeXt-Tiny as a representative convolutional neural network [7] and ViT-B/16 as a transformer-based model [3]. Models are trained using binary cross-entropy for ChestX-ray14 and standard cross-entropy for BDD100K. Model selection is based on validation performance and the best checkpoint is frozen prior to post-hoc calibration. This setup isolates calibration effects from architectural and training differences.

Post-hoc calibration is then performed with all network weights fixed. Calibration parameters are learned on stratified validation subsets of varying size by minimizing the NLL. For each calibration method and subset size, calibrated confidence estimates are evaluated on the held-out test set using global and region-restricted calibration metrics. Experiments are repeated across multiple random seeds and results are reported as mean and standard deviation.

This study assumes in-distribution test data and does not consider distribution shift. Calibration is performed post-hoc without modifying model training and decision-critical regions are defined using fixed confidence thresholds rather than application-specific cost functions. This operationalization reflects deployment settings where predictions below τ are abstained from or manually reviewed.

4.4 Metrics

Calibration quality is evaluated using ECE, Local Expected Calibration Error (Local-ECE) and NLL as standard metrics in neural network calibration [5]. For multi-class classification, the maximum calibrated class probability is interpreted as correctness probability. For TS and ETS, this corresponds to the maximum calibrated class probability (multi-class) or the sigmoid probability assigned to a label (multi-label). For Energy-based Calibration, the logistic mapping directly yields this correctness confidence. All methods are evaluated using correctness-based confidence for comparability. All metrics are computed on binary correctness labels. For multi-label classification, correctness is defined per image-label pair, i.e., a prediction is considered correct if the predicted binary label matches the ground-truth annotation for that label. In this setting, entropy and energy are computed independently per label using the corresponding logit to ensure consistency with label-wise correctness evaluation.

Expected Calibration Error. ECE measures global miscalibration by comparing predicted confidence and empirical accuracy across M confidence bins:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|. \quad (7)$$

Here, B_m denotes the set of samples whose predicted confidence falls into bin m , $|B_m|$ is the number of samples in that bin, n is the total number of samples, $\text{acc}(B_m)$ is the empirical accuracy in bin m and $\text{conf}(B_m)$ is the average predicted confidence in that bin. Lower ECE indicates better alignment between predicted probabilities and actual correctness. We use $M = 15$ equally spaced confidence bins over the interval $[0, 1]$.

Local Expected Calibration Error. To assess calibration in decision-critical settings, we compute Local-ECE by applying equation 7 to the subset of predictions whose confidence lies in the interval $[\tau, 1]$. Concretely, only samples with confidence $\geq \tau$ are included when forming the confidence bins and the normalization constant n in equation 7 is replaced by n_τ , the number of retained high-confidence predictions. Accordingly, each bin B_m is replaced by $B_m^\tau = \{x \in B_m \mid \text{conf}(x) \geq \tau\}$, and $\text{acc}(\cdot)$ and $\text{conf}(\cdot)$ are computed on these restricted bins. Decision-critical regions are defined as $[\tau, 1]$ with $\tau \in \{0.8, 0.9\}$.

Negative Log-Likelihood. NLL evaluates probabilistic prediction quality as a proper scoring rule:

$$\text{NLL} = -\frac{1}{n} \sum_{i=1}^n \log p(y_i \mid x_i), \quad (8)$$

where $p(y_i \mid x_i)$ denotes the predicted probability assigned to the true label y_i . For correctness-based evaluation, this corresponds to the confidence assigned to the predicted class when the prediction is correct and to one minus that confidence otherwise. Lower NLL indicates better probabilistic consistency and penalizes overconfident errors. This evaluation NLL differs from the task-specific NLL used for calibration parameter learning in Section 3.

Experiments are repeated across 10 independent random seeds. Results are reported as mean and standard deviation.

GenAI Usage Disclosure

GPT-5.2 was used solely for language assistance, including grammar correction, minor rephrasing and structural refinement. It had no influence on the scientific reasoning, interpretation or conclusions. All content was thoroughly reviewed and revised by the authors, who bear full responsibility for the final manuscript.

References

- [1] Sergio A. Balanyà, Juan Maroñas, and Daniel Ramos. 2022. Adaptive Temperature Scaling for Robust Calibration of Deep Neural Networks. *Neural Networks* (2022). <https://arxiv.org/abs/2208.00461> arXiv:2208.00461.
- [2] Gregory Canal, Vladimir Leung, John J. Guerrerio, Philip Sage, and I-Jeng Wang. 2024. Decision-Driven Calibration for Cost-Sensitive Uncertainty Quantification. In *Bayesian Decision-making and Uncertainty Workshop at NeurIPS 2024*.
- [3] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*.
- [4] Biraja Ghoshal and Allan Tucker. 2021. On Cost-Sensitive Calibrated Uncertainty in Deep Learning: An Application on COVID-19 Detection. In *2021 IEEE 34th International Symposium on Computer-Based Medical Systems (CBMS)*. IEEE, 503–509. <https://doi.org/10.1109/CBMS52027.2021.00100>
- [5] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 70)*. PMLR, 1321–1330. <https://arxiv.org/abs/1706.04599>
- [6] Mijoo Kim and Junseok Kwon. 2024. Uncertainty Calibration with Energy Based Instance-wise Scaling in the Wild Dataset. In *Proceedings of the Asian Conference on Computer Vision (ACCV)*. <https://arxiv.org/abs/2407.12330> arXiv:2407.12330.
- [7] Zhuang Liu, Hanzhi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. 2022. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [8] Stephen J. McKenna and Jacob Carse. 2024. Calibrating Where It Matters: Constrained Temperature Scaling. In *NeurIPS 2023 Workshop on Bayesian Decision-making and Uncertainty*. <https://arxiv.org/abs/2406.11456> arXiv:2406.11456.
- [9] Matthias Minderer, Josip Djolonga, Rob Romijnders, Frances Hubis, Xiaohua Zhai, Neil Houlsby, Dustin Tran, and Mario Lucic. 2021. Revisiting the Calibration of Modern Neural Networks. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc. 35th Conference on Neural Information Processing Systems (NeurIPS 2021).
- [10] Azadeh Sadat Mozafari, Hugo Siqueira Gomes, Wilson Leão, Steeven Janny, and Christian Gagné. 2019. Attended Temperature Scaling: A Practical Approach for Calibrating Deep Neural Networks. *arXiv preprint arXiv:1810.11586* (2019). <https://arxiv.org/abs/1810.11586>
- [11] Rahul Rahaman and Alexandre H. Thiery. 2021. Uncertainty Quantification and Deep Ensembles. *arXiv preprint arXiv:2007.08792* (2021). <https://arxiv.org/abs/2007.08792>
- [12] Pranav Rajpurkar, Jeremy Irvin, Kaylie Zhu, Brandon Yang, Hershel Mehta, Tony Duan, Daisy Ding, Aarti Bagul, Robyn L. Ball, Curtis Langlotz, Katie Shpanskaya, Matthew P. Lungren, and Andrew Y. Ng. 2017. CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning. *arXiv preprint arXiv:1711.05225* (2017). arXiv:1711.05225
- [13] Melanie Roschewitz, Raghav Mehta, Fabio De Sousa Ribeiro, and Ben Glocker. 2025. Where are we with calibration under dataset shift in image classification? *Transactions on Machine Learning Research* (2025). arXiv:2507.07780 <https://openreview.net/forum?id=1NYKXIRU2H> Published in TMLR (10/2025). Reviewed on OpenReview.
- [14] Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. 2017. ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 3462–3471. <https://arxiv.org/abs/1705.02315>
- [15] Mochen Yang and Xuan Bi. 2025. Cost-Aware Calibration of Classifiers. *INFORMS Journal on Data Science* 4, 2 (2025), 101–113. <https://doi.org/10.1287/ijds.2024.0038>
- [16] Suorong Yang, Weikang Xiao, Mengchen Zhang, Suhan Guo, Jian Zhao, and Furoo Shen. 2023. Image Data Augmentation for Deep Learning: A Survey. *arXiv preprint arXiv:2204.08610* (2023). arXiv:2204.08610 [cs.CV]
- [17] Fisher Yu, Haofeng Chen, Xin Wang, Wenqi Xian, Yingying Chen, Fangchen Liu, Vashisht Madhavan, and Trevor Darrell. 2020. BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2636–2645. <https://arxiv.org/abs/1805.04687>

A Supplementary Materials

A.1 Dataset Details

Table 2: Datasets used in this study.

Dataset	Domain	Task	Classes	Samples	Split (Train/Val/Test)
ChestX-ray14 [14]	Medical	Multi-label disease classification	14 thoracic diseases (Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, Hernia)	112,120 images (30,805 patients)	Official patient-level split
BDD100K [17]	Driving	Image-level scene classification	6 scene types (residential, highway, city street, parking lot, gas stations, tunnel)	100,000 images	70K / 10K / 20K

A.1.1 BDD100K Image-Level Attributes. The dataset provides six scene types (residential, highway, city street, parking lot, gas stations, tunnel), six weather conditions (clear, partly cloudy, overcast, rainy, snowy, foggy) and three times of day (dawn/dusk, daytime, night) [17].

A.2 Additional Metric Definitions

Table 3: Overview of calibration metrics

Metric	What it measures	Why it matters
ECE	Weighted average discrepancy between predicted confidence and empirical accuracy across confidence bins	Quantifies global calibration quality
Local-ECE	ECE restricted to predictions above a predefined confidence threshold	Assesses calibration in decision-critical settings
NLL	Negative log-probability of prediction correctness (proper scoring rule)	Penalizes miscalibration and overconfident errors